FRICTION — BRAKE PADS INVESTIGATION

Detailed Lesson Plan · Planrs Academy Science Lab

Session	Duration	Type	Subject	Depth
Session 1	**30 Minutes**	**Hands-on Lab**	**Science / EVS**	**Beginner**

1. LESSON OVERVIEW

What this session is all about

This session introduces learners to the concept of **FRICTION** through the exciting, real-world context of brake pads on bicycles, autos, and cars. Learners investigate how different surfaces create different amounts of friction, discover why rough surfaces slow things down faster than smooth ones, and connect the big idea to everyday Indian life. The session is **activity-first, curiosity-driven**, and built around a simple hands-on investigation.

■ The Golden Idea: **Rough surfaces grip more** → **More friction** → **Stops faster!** This is exactly how brake pads work to keep us safe on bicycles, auto-rickshaws, school buses, and scooters.

Prior Knowledge Assumed	Forward Curriculum Links
✥ Objects can move and stop	✥ Push and Pull forces (Class 3–4)
✥ Awareness of bicycles & vehicles	✥ Friction chapter in EVS (Class 5)
✥ No prior physics knowledge required	✥ Force and Pressure — NCERT Class 8
✥ Basic motor skills for sliding activity	✥ Laws of Motion — NCERT Class 9

2. LEARNING OBJECTIVES

By the end of this session, learners will be able to:

LO	Learning Objective	Bloom's Level	Measured By
LO 1	Name friction as the force that slows things down when two surfaces rub together	Remember	Exit ticket / vocab quiz
LO 2	Explain why rough surfaces create more friction than smooth surfaces	Understand	Discussion questions
LO 3	Predict which surface creates more friction before testing, and compare to result	Apply	Observation worksheet
LO 4	Give at least one real-life Indian example of friction being helpful	Apply / Analyse	Think & Share discussion
LO 5	Participate in the surface-comparison investigation and record observations	Analyse	Worksheet completion

■ Success Indicator: A learner has met all objectives when they can say — "Friction happens when things rub together. Rough surfaces make more friction. Brake pads are rough so they stop the wheel. Without friction we would slip!"

3. KEY CONCEPTS WITH INDIAN EXAMPLES

Core science ideas for this session

Concept 1 — What is Friction?

Friction is the **force that happens when two things rub against each other**. It slows moving things down or stops them completely. Core idea: **Friction = Rubbing = Slowing Down.**

Indian Example	What is Happening
Rubbing hands together on a cold Delhi winter morning	Friction between palms generates heat — hands feel warm!
Cricket ball rolling on the pitch and slowing down	Friction between ball and ground surface slows the ball
Sliding on a polished marble temple floor vs. rough cemented courtyard	Marble (smooth) = less friction; Cement (rough) = more friction
Dosa sliding on a well-oiled tawa vs. a dry pan	Oil reduces friction — dosa slides easily; dry pan grips more

Concept 2 — Rough vs. Smooth Surfaces

■ ROUGH — More Friction — Stops Faster	■ SMOOTH — Less Friction — Keeps Moving
■ Sandpaper — very rough, lots of friction	■ Glass surface of a mobile screen
■ Rubber shoe sole — grips the floor	■■ Polished marble temple floor
■ Cycle tyre tread — grips the road	■ Inside of a steel tumbler (gilaas)
■ Coir doormat at Indian home entrance	■ Smooth park slide surface
■ Cricket pitch surface	■ Wet bathroom floor (monsoon danger!)
■ Auto-rickshaw tyre on a market road	■ Plastic sheet / aluminium foil

Concept 3 — Brake Pads and How They Work

Brake pads are **rough pads inside wheel mechanisms**. When the brake lever is squeezed, the rough pad presses against the spinning wheel. This creates a large amount of friction that slows the wheel and stops the vehicle.

1. Squeeze brake handle with your hand
2. Rough brake pad presses against the spinning wheel
3. LOTS of friction is created → Wheel slows → Bicycle stops → You are SAFE!

Indian Vehicle	Brake Type	Why Friction Matters
Hero / Atlas Bicycle	Rubber rim brake pad	Presses wheel rim; stops in a short distance
Auto-rickshaw (Three-wheeler)	Drum / disc brake	Stops safely at school gates and traffic signals
School Bus (Yellow)	Air / hydraulic drum brake	Heavy vehicle needs very strong friction to stop
Honda Activa / Hero Splendor Scooter	Disc brake (front)	Parents' daily commute — friction keeps family safe

Concept 4 — Friction is Helpful (and Sometimes Challenging!)

■ Helpful Friction — Everyday Examples

- ■ Chappal grip on a wet monsoon floor
- ■ Writing with a pencil — graphite sticks to paper
- ■ Cricket fielder gripping the ball
- ■ Matchstick on matchbox — lights with friction
- ■ Tyre grip stops school bus from skidding
- ■ Brake pads stopping bicycles safely

■■ Challenging Friction — When It Works Against Us

- ■■ Heavy almirah hard to push across the floor
- ■■ Machinery parts wear out over time
- ■■ Dragging heavy school bag damages it
- ■■ Brake pads wear thin and need replacing
- ■■ Increased fuel needed to overcome road friction
- ■■ Moving furniture requires more effort indoors

4. SESSION OUTLINE WITH TIMING

Total Duration: 30 Minutes · 5 Phases

1 ■ Hook & Wonder 0:00 – 0:05 (5 mins)

Purpose: Spark curiosity and activate prior knowledge through a compelling question.

Opening Question	Ask learners: "Has anyone ever been on a bicycle and pressed the brakes? What happened? WHY did it stop?" Allow 2–3 responses. Accept all answers enthusiastically.
Spot the Friction!	Show (or describe) three scenes: (1) A child pressing bicycle brakes on a busy Indian street. (2) Two hands rubbing together. (3) A cricket ball rolling on the pitch and slowing down. Ask: "What is the SAME in all three pictures?" Guide toward: rubbing, slowing, stopping.
Introduce the Word	Say: "The word for this is FRICTION! Say it with me — FRIK-SHUN!" Write it on the board. "Today we are going to be FRICTION DETECTIVES!"

2 ■ Concept Building 0:05 – 0:12 (7 mins)

Purpose: Build the core understanding — what friction is, rough vs. smooth, and how brake pads work.

Step 1 · 2 min	Learners rub their palms together at their desks. Ask: "What do you feel?" (Warmth.) Explain: "When two things rub, friction is created. Friction slows things down." Show a rough sandpaper piece and a smooth plastic sheet side by side.
Step 2 · 2 min	Rough vs Smooth: "Rough surfaces make MORE friction. Smooth surfaces make LESS friction." Indian examples: Rubber chappal on a wet floor (slips = less friction) vs. rubber-soled school shoes (grips = more friction). Students call out rough/smooth things from their classroom or home.
Step 3 · 3 min	Brake Pads Explained: Show a simple diagram of a bicycle wheel and brake pad. Walk through the 3-step braking process. Ask: "What would happen if the brake pad was SMOOTH like glass? Would the bike stop?" (No!) Discuss Hero cycle and Atlas cycle as familiar references.

3 ■ Hands-On Investigation 0:12 – 0:22 (10 mins)

Purpose: Learners experience friction directly through a structured investigation.

Investigation: "Which Surface Stops the Block Fastest?"

Item	Qty	Indian Source / Note
Wooden block or small plastic tiffin box (dabbawala dabba)	1	Any tiffin box or geometry box from school bag
Sandpaper sheet	1 piece	Available at any hardware shop (bhandaar)
Newspaper page	1 sheet	Any daily newspaper — smooth-ish surface
Aluminium foil / polythene sheet	1 piece	Kitchen foil — very smooth, low friction
Old cloth / jute bag piece	1 piece	Cut from old dupatta or jute shopping bag
Ramp (school textbook or wooden plank, ~30cm)	1	Stack 2–3 textbooks at an angle
Pencil / sticky note for marking stop point	1–2	Mark where block stops on each surface
Observation Worksheet (printed)	1 per learner	See Section 7 of this document

PREDICT (2 min)	Learners look at the 4 surfaces. They circle or write their prediction on the worksheet: "I think the ________ surface will stop the block fastest because it looks ________."
TEST (5 min)	Teacher demonstrates first: Place surface at base of ramp. Slide block down gently. Mark the stopping point. Then student pairs try each surface. Ramp height stays constant throughout all tests (fair test).
OBSERVE & RECORD (3 min)	Learners draw or write: "On the ________ surface, the block stopped: NEAR / FAR (circle one). This surface is: ROUGH / SMOOTH (circle one)."

■■ Safety Note: Use a LOW ramp only. Block must be lightweight. Adult/teacher pours no hot or sharp materials. Remind learners: we slide the block gently — no throwing!

4 ■ Discuss & Connect

0:22 – 0:27 (5 mins)

Purpose: Make sense of findings; connect to real life; reinforce vocabulary.

Discussion Question	Expected Response / Teaching Note
Q1: Which surface stopped the block fastest? Why?	Sandpaper — it is the roughest, so it creates the most friction. Guide: "Rough = more grip = stops faster."

Discussion Question	Expected Response / Teaching Note
Q2: Which surface is most like a brake pad — sandpaper or aluminium foil? Why?	Sandpaper — it is rough like a brake pad. Brake pads are designed rough ON PURPOSE so they create lots of friction to stop the wheel.
Q3: Where does friction keep YOU safe at home or on your street?	Accept all answers. Prompt if needed: Chappal sole, school shoe on stairs, auto-rickshaw brakes at school gate, cricket bat grip, doormat at home entrance.

5 ■ Wrap-Up & Recap

0:27 – 0:30 (3 mins)

3 Big Things We Learned Today:

1	Friction happens when two surfaces rub together
2	Rough surfaces make MORE friction — they stop things faster
3	Brake pads are rough on purpose — they use friction to stop wheels!

■ Exit Ticket — Thumbs Up / Down: Show each object and ask learners to hold THUMBS UP (more friction) or THUMBS DOWN (less friction). (1) Rough Sandpaper → UP | (2) Smooth Plastic Sheet → DOWN | (3) Rubber Chappal Sole → UP

■ Take-Home Challenge — Friction Detective Homework: Find 3 things at home that use friction to keep you safe. Draw them and bring the drawings next session! Prompts: doormat, rubber handle on a vessel, slippers, bicycle brakes, eraser on paper.

5. VOCABULARY GLOSSARY

Key words introduced in this session

FRICTION

The force that happens when two surfaces rub. Friction slows things down or stops them.

Hindi: Ghurshan (■■■■■)

Example: Hands rubbing together feel warm

ROUGH

A surface with bumps and texture — creates MORE friction.

Example: Sandpaper, coir mat, tyre tread

SMOOTH

A surface that is flat and slippery — creates LESS friction.

Example: Marble floor, glass, plastic sheet

BRAKE PAD

A rough pad inside a wheel that presses the wheel to create friction and stop it.

Example: Bicycle brake, auto-rickshaw brake

GRIP

When a surface holds onto another without slipping — made possible by friction.

Example: Rubber sole gripping wet floor

SURFACE

The outer face of an object — what you can touch. Can be rough or smooth.

Example: Sandpaper surface vs. glass surface

PREDICT

Your best guess about what will happen BEFORE you test it.

Example: I predict sandpaper stops the block fastest

OBSERVE

To look carefully and notice what is happening during the test.

Example: I observe the block stopped near the ramp

CONCLUDE

To decide what you learned from your investigation results.

Example: Rough surfaces have more friction

6. COMMON MISCONCEPTIONS

Watch for these — and how to address them

Misconception	Reality	Teaching Tip
"Friction only happens when things are moving."	Friction also prevents stationary objects from moving (static friction).	Place a book on a tilted surface — friction stops it from sliding even when still.

Misconception	Reality	Teaching Tip
"Smooth surfaces have NO friction at all."	All surfaces have some friction; smooth ones have LESS, not zero.	Say: "Even the marble floor has a little friction — but much less than sandpaper."
"Friction is always bad — it causes problems."	Without friction we could not walk, write, or stop safely. It is essential.	Build helpful-friction examples very strongly before any challenging-friction examples.
"Brake pads are inside the engine of the vehicle."	Brake pads act on the WHEELS, not the engine.	Use a clear diagram: show brake lever → cable → pad pressing the wheel rim.
"Heavier things do not need friction to stop."	Heavier things actually need MORE friction to stop — they carry more energy.	Say: "A heavy school bus takes longer to stop than a bicycle for this reason."

7. ASSESSMENT RUBRIC

For educator evaluation — 5 criteria, 4-point scale

Criteria	Excellent (4)	Good (3)	Developing (2)	Needs Support (1)
Concept Understanding	Clearly explains friction, rough/smooth, and brake pads using correct terms	Explains with minor gaps or errors	Partially explains — some confusion between rough and smooth	Cannot yet explain the concept
Observation Skills	Accurately describes block movement with detail on all 4 surfaces	Describes movement with some detail (3 of 4 surfaces)	Notes some movement but lacks specific detail	Cannot identify or record movement
Prediction & Testing	Makes logical prediction; links result to science explanation	Makes a prediction; partially explains why result happened	Makes a guess; difficulty linking result to science	No prediction attempted
Real-life Connections	Names 2+ Indian examples and explains HOW friction works in each	Names 1 Indian example with explanation	Names an example but cannot explain the friction connection	Cannot name a real-life example

Criteria	Excellent (4)	Good (3)	Developing (2)	Needs Support (1)
Worksheet / Drawing	Labelled diagram with arrows, colours, surface names, and friction terms	Diagram with arrows and some labels	Basic drawing with minimal labels	Incomplete or missing

8. STUDENT OBSERVATION WORKSHEET

Printable — to be completed during the investigation

Name: ________________________________ Date: ____________________

■ BEFORE THE EXPERIMENT — My Prediction

I think the ______________ surface will stop the block fastest because it looks ______________.

■ Sandpaper ■ Newspaper ■ Aluminium Foil ■ Cloth

■ OBSERVATION TABLE — Record Your Results

Surface	Rough or Smooth?	Block stopped: Near / Far?	More or Less Friction?
■ Sandpaper	Rough / Smooth (circle)	Near the ramp / Far away (circle)	More / Less (circle)
■ Newspaper	Rough / Smooth (circle)	Near the ramp / Far away (circle)	More / Less (circle)
■ Aluminium Foil	Rough / Smooth (circle)	Near the ramp / Far away (circle)	More / Less (circle)
■ Cloth	Rough / Smooth (circle)	Near the ramp / Far away (circle)	More / Less (circle)

■ What I Learned

A convection current — wait, wrong topic! A convection — no! **FRICTION** is:

____

The surface with the MOST friction was: ____________________________

The surface with the LEAST friction was: ____________________________

One place in India where friction keeps me safe: ____________________________

✏■ Draw It! — Label your diagram

(Draw a glass of water — wait, wrong lesson! Draw the ramp and block below. Show which surface stops the block fastest with a red arrow. Which surface lets it slide the farthest? Use a green arrow.)

9. INDIAN CONTEXT REFERENCE BANK

To be woven into all content delivery

Category	Examples	Friction Connection
Vehicles	Hero/Atlas bicycle, auto-rickshaw, school bus, Honda Activa, Hero Splendor, cycle-rickshaw	All use friction-based brake pads to slow and stop
Footwear	Rubber chappals, Bata school shoes, Paragon slippers, hawai chappals	Rough soles grip floor — more friction = safer walking
Household Objects	Coir doormat, jute bag, steel tiffin box, wooden belan, matchstick on matchbox	Rough surfaces grip; smooth surfaces slide; matchstick uses friction to light
Sports	Cricket ball on pitch, cricket bat rubber grip, kabaddi on mud, Sports Day running track	Friction slows ball; grip prevents bat slipping; shoes grip track
Seasonal Context	Monsoon wet roads, Delhi winter hand-rubbing, dusty summer village roads	Wet surfaces reduce friction (danger!); rubbing hands creates warmth via friction
Food / Kitchen	Dosa on oiled tawa vs. dry pan, chapati and belan, grinding stone (sil-batta)	Oil reduces friction; rough surfaces grip and hold during grinding

10. SUCCESS CRITERIA

Three levels — Learner, Content, and Delivery

■■■ Learner Success

- ✥ Can name friction in one simple sentence
- ✥ Correctly identifies rough vs. smooth from images (80%+ accuracy)

✦ Explains WHY brake pads are rough (not just "so the bike stops")

✦ Names one Indian friction example without prompting

✦ Completed both prediction AND observation on the worksheet

✦ Scores 70%+ on the 5-question post-session quiz

■ Content Quality

✦ All 5 Learning Objectives addressed across outputs

✦ Minimum 3 authentic Indian examples in every major output

✦ All key vocabulary introduced, defined, and reinforced (3x)

✦ Investigation uses locally available materials only

✦ Zero factual errors (verified against NCERT/CBSE science)

✦ Session delivers fully within 30 minutes

■ Delivery Quality

✦ Learner stays engaged throughout (no "blank stare" moments)

✦ Indian character illustrations — diverse skin tones, school uniforms

✦ All slide text minimum 28pt; all print text minimum 14pt

✦ Worksheet printable in black-and-white with full clarity

✦ Video narration at 110-130 words per minute

✦ Planrs Academy branding on all outputs

Planrs Academy · Science Lab · Friction — Brake Pads Investigation · Session 1

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"Learning that feels like play — for every Indian child."